VIETNAM NATIONAL UNIVERSITY - HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Programming for Artificial Intelligence and Data Science (CO3135)

Assignment Submission

House Price Prediction

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1 Group Information & Member Roles

The registered group will work together on all assignments in this course. The responsibilities for this phase are divided as follows:

- **Member 1: Võ Nhật Tân**
  - Role: xxx.
- **Member 2: Ngân Phạm**
  - Role: xxx.
- **Member 3: Minh Thiện**
  - Role: xxx.
- **Member 4: Chi Phan**
  - Role: xxx.

2 Chosen Data & Justification

The chosen dataset provides a large number of real-estate listings with a clear numeric target and a wide range of predictors, which suits the house price prediction task for the Vietnamese market.

Dataset *Tinix Vietnam Real Estate Listings 2025–2026* [Tin26]

Overview According to the published dataset card, the collection contains exactly 3,500,744 sale and rental listings published between June 2025 and March 2026, after a strict property-type filter, and covers all 63 provinces and cities of Vietnam. The listings are distributed as ten Parquet shards, are written primarily in Vietnamese, and are released under the CC BY-NC 4.0 licence for non-commercial academic use.

Justification The dataset contains over 3.5 million records, which supports stable exploratory analysis and supervised learning across regions and property types. Its schema separates a numeric target (`price`) from predictors describing location, property category, size and layout, orientation, textual content, and listing time. National coverage makes it possible to study regional price patterns and to train models that generalise beyond a single city, and the publication timestamp allows chronological train/validation/test splits, which are preferable to random splits for a market that changes over time. The publisher also lists real-estate price prediction among the intended use cases.

Limitations The published definitions indicate that `price` may be undisclosed (`null`), that some structural fields are often incomplete for particular property types, and that the collection includes both sale and rental listings, with no variable that distinguishes them. These issues are examined in Section 4, where an analytical subset of priced listings is constructed for subsequent analysis and model training.

Table 1 reproduces the data schema from the published dataset card. The descriptions are taken from that page and are not reinterpreted here.

Table 1: Feature meanings from the published dataset card [Tin26].

Column	Type	Description (published)
`name`	string	Listing title (HTML stripped, phone numbers redacted)
`description`	string	Full listing description (HTML stripped, phone numbers redacted)
`property_type_name`	string	Property category (e.g., Căn hộ chung cư, Nhà, Đất, ...)
`province_name`	string	Province / City name (63 provinces)
`district_name`	string	District (Quận / Huyện)
`ward_name`	string	Ward (Phường / Xã)
`street_name`	string	Street name
`project_name`	string	Real estate development project name (if applicable)
`price`	float64	Listed price in VND (Vietnamese Dong). `null` if not disclosed.
`area`	float64	Total floor/land area in m^2
`floor_count`	float64	Number of floors in the building
`frontage_width`	float64	Frontage width in meters
`house_depth`	float64	Depth of the property in meters
`road_width`	float64	Width of the road in front of the property (meters)
`bedroom_count`	float64	Number of bedrooms
`bathroom_count`	float64	Number of bathrooms
`house_direction`	string	Cardinal facing direction (Đông, Tây, Nam, Bắc, Đông-Bắc, ...)
`balcony_direction`	string	Cardinal facing direction of the balcony
`published_at`	string	ISO 8601 datetime the listing was published

3 Work Progress Overview

4 Exploratory Data Analysis (EDA)

This section presents the data quality checks, the construction of the analytical dataset, and the price analysis. The work was carried out in two notebooks: the first covers data quality and cleaning, and the second covers the analysis of price and area. Each listing is treated as a property for sale, and `price` is the asking price in VND. The results therefore cover only listings that show a price, and some rental listings may remain (see Section 2).

4.1 Data Quality Analysis

4.1.1 Schema and Data Types

The raw data contain 3,500,744 listings and the 19 variables described in Table 1. The schema check confirmed that all expected columns are present and that there are no additional columns. The data types are as follows:

- Numeric (8 variables): `price`, `area`, `floor_count`, `frontage_width`, `house_depth`, `road_width`, `bedroom_count`, `bathroom_count`.
- Text (11 variables): all remaining variables.

To reduce memory usage, six categorical columns were converted to the categorical type, which reduced the memory footprint from 6.533 GB to 5.334 GB. A conservative cleaning strategy was adopted: quality issues were flagged and all rows were kept in the full dataset, while a separate subset of priced listings was built afterwards (Section 4.2).

4.1.2 Missing-Value Analysis

The null-value scan showed missing entries in 14 of the 19 variables (Table 2). The main findings are:

- `price`: 218,494 missing values (6.24%), and `area`: 4 missing values.
- Structural attributes have high missing rates, ranging from 68.96% for `house_direction` to 98.04% for `house_depth`.
- `name`, `description`, `property_type_name`, `province_name`, and `published_at` have no missing values.

Table 2: Missing values by variable in the raw data ($N = 3{,}500{,}744$).

Variable	Missing (n)	Missing (%)
`house_depth`	3,432,288	98.04
`balcony_direction`	2,933,342	83.79
`road_width`	2,830,873	80.86
`floor_count`	2,691,982	76.90
`project_name`	2,499,891	71.41
`house_direction`	2,414,214	68.96
`frontage_width`	1,726,424	49.32
`bathroom_count`	1,720,974	49.16
`bedroom_count`	1,600,361	45.71
`street_name`	1,376,860	39.33
`ward_name`	496,716	14.19
`price`	218,494	6.24
`district_name`	101,933	2.91
`area`	4	<0.01

The missing rate was also computed for each property type.

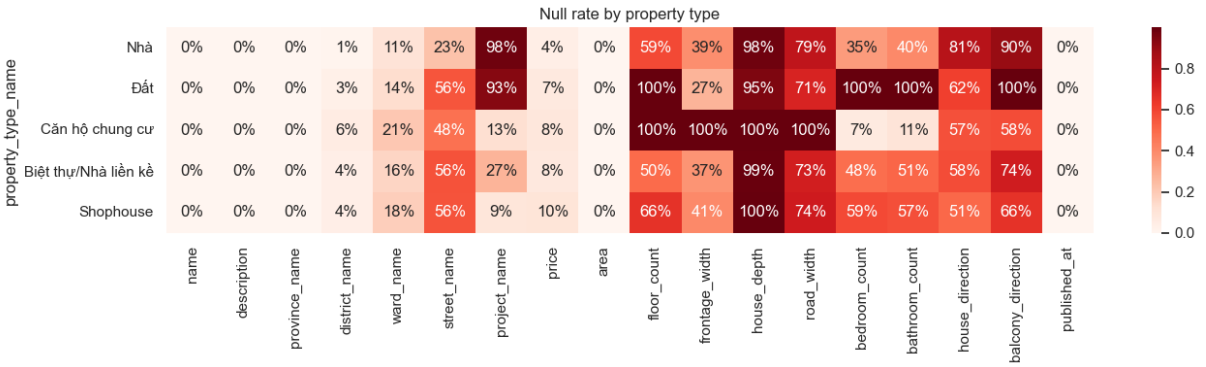


Figure 1: Null rate of each variable by property type.

Figure 1 shows the null rate of each variable for the five property types. The pattern differs across types, which indicates that part of the missingness is related to the nature of the property:

- Apartments (*Căn hộ chung cư*): `floor_count`, `frontage_width`, `house_depth`, and `road_width` are missing in 100% of the listings.
- Land (*Đất*): `floor_count`, `bedroom_count`, `bathroom_count`, and `balcony_direction` are missing in 100% of the listings.
- `project_name` is missing in 98% of houses (*Nhà*) and 93% of land listings, compared with 13% of apartments. A missing `project_name` often indicates that the listing does not belong to a named development project.
- `price` is missing in 4% to 10% of the listings of each type.

Rows with missing values were retained in the dataset, and listings without a valid price are excluded later from the price analysis.

4.1.3 Validity Checks

Validity checks were applied to the numeric variables:

- `price`: 57,320 listings have a price equal to zero, and no listing has a negative price.
- `area`: no listing has an area less than or equal to zero.
- `floor_count`, `bedroom_count`, `bathroom_count`: no negative values.

Listings without a valid price or area represent 7.88% of the raw data. This share ranges from 5.2% (*Nhà*) to 13.6% (*Shophouse*) across the five property types, and from 4.1% (Khánh Hòa and Lâm Đồng) to 18.8% (Hòa Bình) among the 15 provinces with the most listings. These listings are therefore spread across property types and provinces, and no group accounts for most of them.

4.1.4 Temporal, Geographic and Direction Checks

The following checks were applied to the time, location, and direction variables:

- `published_at`: no missing values in the raw data, but 22,024 values could not be converted to a timestamp (invalid values were coerced to missing). The parsed values span 1 June 2025 to 30 March 2026.
- Geography: the data contain 63 provinces and 627 districts. A total of 15 district names appear in more than one province, so `district_name` should be used together with `province_name`.
- Direction: `house_direction` and `balcony_direction` are free text, with 2,414,214 and 2,933,342 missing values, respectively. The same direction is sometimes written in two notations (for example, *Đông Nam* and *Đông - Nam*), and a few entries contain several directions, each occurring at most five times in `house_direction` and at most four times in `balcony_direction`. No standardisation was applied at this stage.

4.1.5 Duplicate Analysis

Listings were flagged as duplicates when they share the same normalised text of `name`, `province_name`, `district_name`, and `ward_name` (lower case, diacritics removed, *đ* mapped to *d*, whitespace collapsed), together with the same `area` and `price`, as an earlier listing. The first occurrence is kept.

```
df["is_dup"] = dup_check.duplicated(keep="first")
```

Under this definition, 865,362 listings (24.72% of the raw data) are duplicates. The ten most repeated combinations occur between 60 and 121 times each, all in Hồ Chí Minh, and eight of the ten share the same title opening (*xuat canh ban gap*, in normalised form). The most repeated combination is a house on Lê Thánh Tôn street (Bến Nghé ward, District 1) of about 43 m^2, priced at 15.5 billion VND, which appears 121 times.

4.1.6 Extreme Value Analysis

The distribution of `price` in the priced, de-duplicated data was plotted on logarithmic axes.

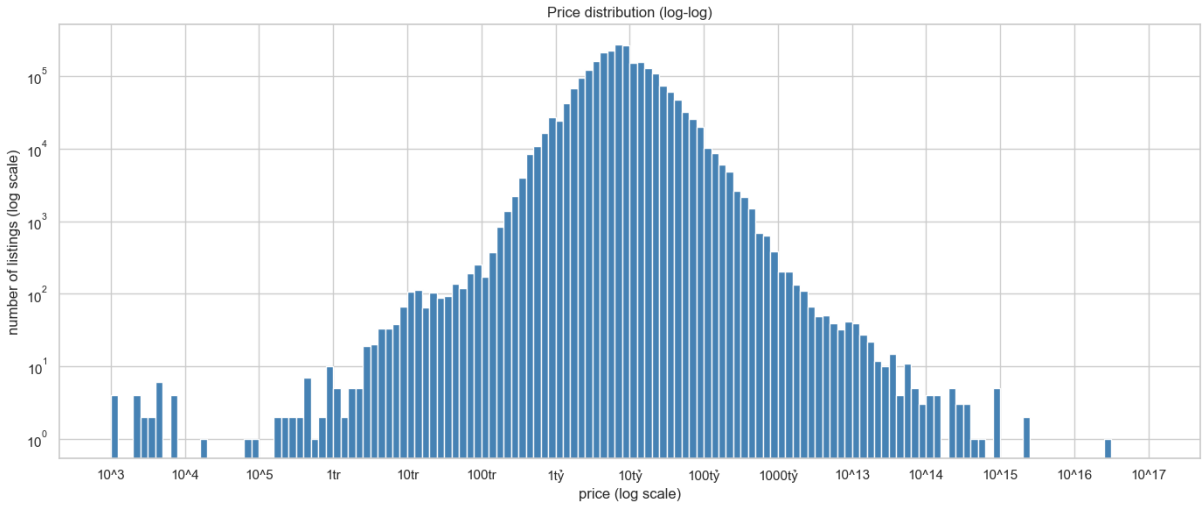


Figure 2: Distribution of asking price after removing invalid prices and duplicates (logarithmic scale on both axes).

Figure 2 shows that most listings are concentrated between about 10^9 and 10^{10} VND, with isolated listings below 10^4 VND and above 10^{15} VND. The ten lowest and ten highest prices were inspected:

- The ten lowest prices are between 1,000 and 3,000 VND. In several of these listings the title states a different price; for example, a listing recorded at 2,000 VND is titled as 1.6 billion VND.
- The ten highest prices range from about 4.49×10^{14} to 2.66×10^{16} VND, and some of these listings have very large areas (for example, 999,999 m^2).

These observations suggest data entry errors, for example in the unit of the price. A range of $[10^8, 10^{12}]$ VND (100 million to 1,000 billion VND) was therefore examined: it would remove 2,574 listings (0.107%), and it was applied as a cleaning step (Section 4.2).

The numeric attributes and the price per square metre were inspected in the same way.

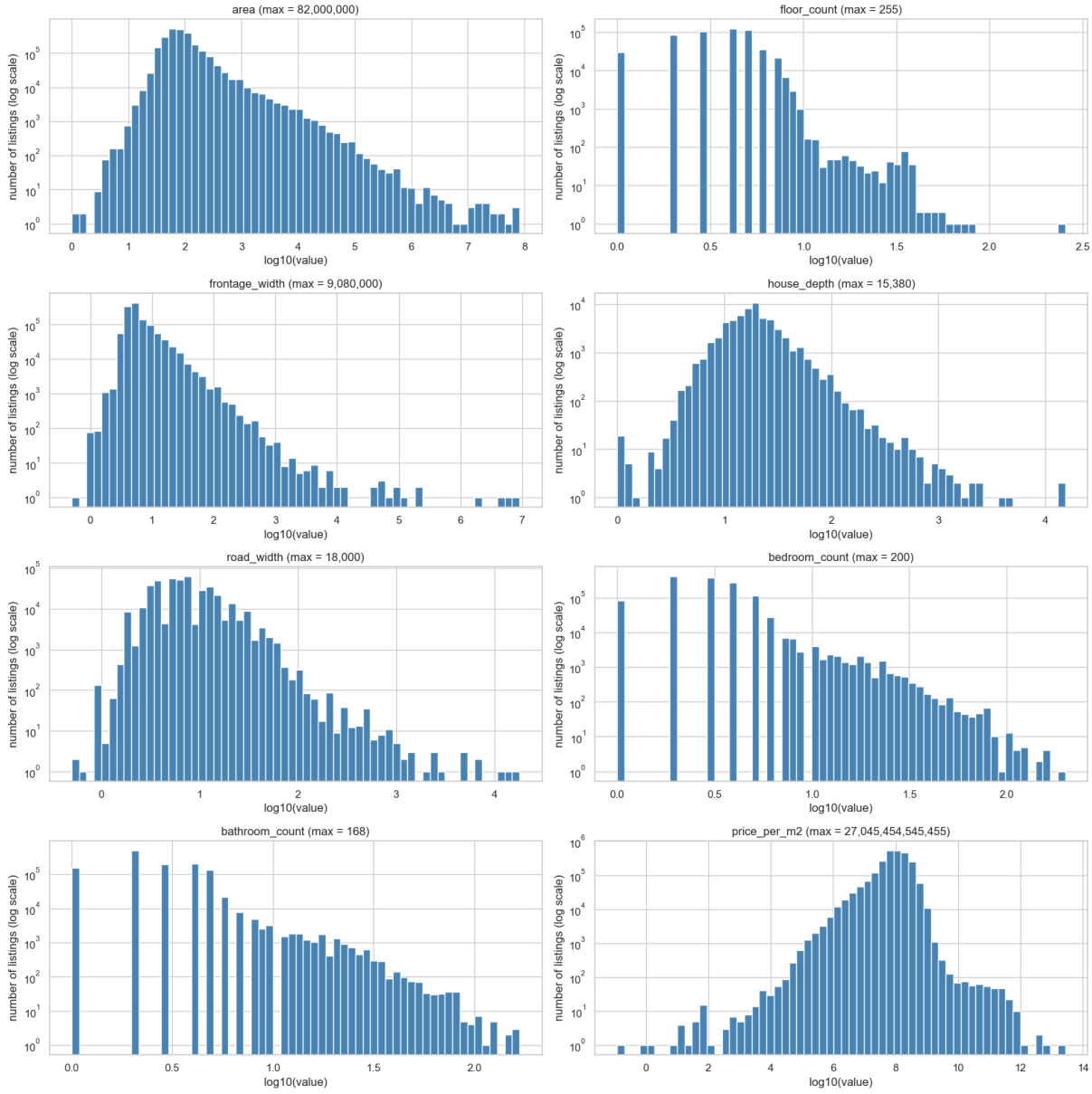


Figure 3: Distributions of the numeric attributes and of the price per square metre ($\log_{10}$ scale; vertical axis in logarithmic scale), before the plausibility limits are applied.

Figure 3 shows that in each panel most listings lie in a narrow range, while sparse tails extend over several orders of magnitude. The maximum values are 82,000,000 m^2 for `area`, 255 for `floor_count`, 9,080,000 m for `frontage_width`, 15,380 m for `house_depth`, 18,000 m for `road_width`, 200 for `bedroom_count`, 168 for `bathroom_count`, and about 2.70×10^{13} VND per m^2 for `price_per_m2`. These values are not plausible for most properties. The separated bars for `floor_count`, `bedroom_count`, and `bathroom_count` reflect that these variables take integer values.

4.2 Analytical Dataset Construction

The analytical dataset (`df_priced`) was built from the full dataset in six steps, applied in the order of Table 4. The first three steps retain listings with a valid price and area, remove the flagged duplicates, and retain prices within $[10^8, 10^{12}]$ VND:

```
df_priced = df[df["price"].gt(0) & df["area"].gt(0)]
df_priced = df_priced[~df.loc[df_priced.index, "is_dup"]]
df_priced = df_priced[df_priced["price"].between(1e8, 1e12)]
```

The remaining steps remove listings without a valid publication date, apply plausibility limits to `area` and the other numeric attributes (Table 3), and remove listings with a price per square metre above 2 billion VND. For `area`, listings outside the limits were removed. For the other six attributes, the out-of-range values were set to missing and the listings were retained. The number of values outside the limits was counted before the area filter was applied. The final dataset contains 2,397,001 listings.

Table 3: Plausibility limits applied to the numeric attributes.

Variable	Valid range	Values outside (n)	Treatment
`area` (m^2)	$[10, 100{,}000]$	876	Row removed
`floor_count`	$[1, 50]$	8	Set to missing
`frontage_width` (m)	$[1, 200]$	761	Set to missing
`house_depth` (m)	$[2, 500]$	70	Set to missing
`road_width` (m)	$[1, 100]$	424	Set to missing
`bedroom_count`	$[1, 50]$	340	Set to missing
`bathroom_count`	$[1, 50]$	267	Set to missing

Table 4: Number of listings after each filtering step.

Step	Rule	Listings (n)	Removed (n)
0	Raw data	3,500,744	—
1	Price > 0 and area > 0	3,224,930	275,814
2	Duplicates removed	2,415,372	809,558
3	Price within $[10^8, 10^{12}]$ VND	2,412,798	2,574
4	Valid publication date	2,398,343	14,455
5	Area within $[10, 100{,}000]$ m^2	2,397,467	876
6	Price per $m^2 \leq 2 \times 10^9$ VND (final)	2,397,001	466

The number removed in step 2 is the difference between the counts before and after the step.

4.3 Price Analysis

All analyses in this subsection use the analytical dataset (`df_priced`).

4.3.1 General Dataset Profile

Table 5 reports descriptive statistics of the numeric variables. The main observations are:

- `price` and `area` are available for all listings, whereas the other attributes are available for fewer listings (from 58,878 for `house_depth` to 1,304,397 for `bedroom_count`).
- The median price is 7.4 billion VND and the mean (1.49×10^{10} VND) is about twice as large, which indicates a right-skewed distribution.
- `area` behaves similarly: the median is 80 m^2, the mean is 217 m^2, and the standard deviation is 1,619 m^2.
- The maximum of each attribute equals the upper limit in Table 3 (for example, 100,000 m^2 for `area` and 200 m for `frontage_width`), so the tails are bounded.

4.3.2 Distribution of Price and Area

The distributions of `price` and `area` were plotted after a base-10 logarithmic transformation.

Table 5: Descriptive statistics of numeric variables in the analytical dataset ($N = 2{,}397{,}001$).

Variable	Count	Mean	Std. dev.	Min	25%	50%	75%	Max
price (VND)	2,397,001	1.49×10^{10}	3.20×10^{10}	1.00×10^{8}	4.05×10^{9}	7.40×10^{9}	1.43×10^{10}	1.00×10^{12}
area (m^2)	2,397,001	217.38	1,619.32	10.00	56.00	80.00	120.00	100,000
floor_count	516,628	3.90	1.76	1	3	4	5	50
frontage_width (m)	1,208,300	7.54	8.97	1.00	4.20	5.00	7.50	200
house_depth (m)	58,878	22.57	20.84	2.00	13.41	18.75	24.64	500
road_width (m)	414,645	9.08	8.62	1.00	4.00	6.00	10.50	100
bedroom_count	1,304,397	3.30	2.34	1	2	3	4	50
bathroom_count	1,220,972	3.05	2.34	1	2	2	4	50

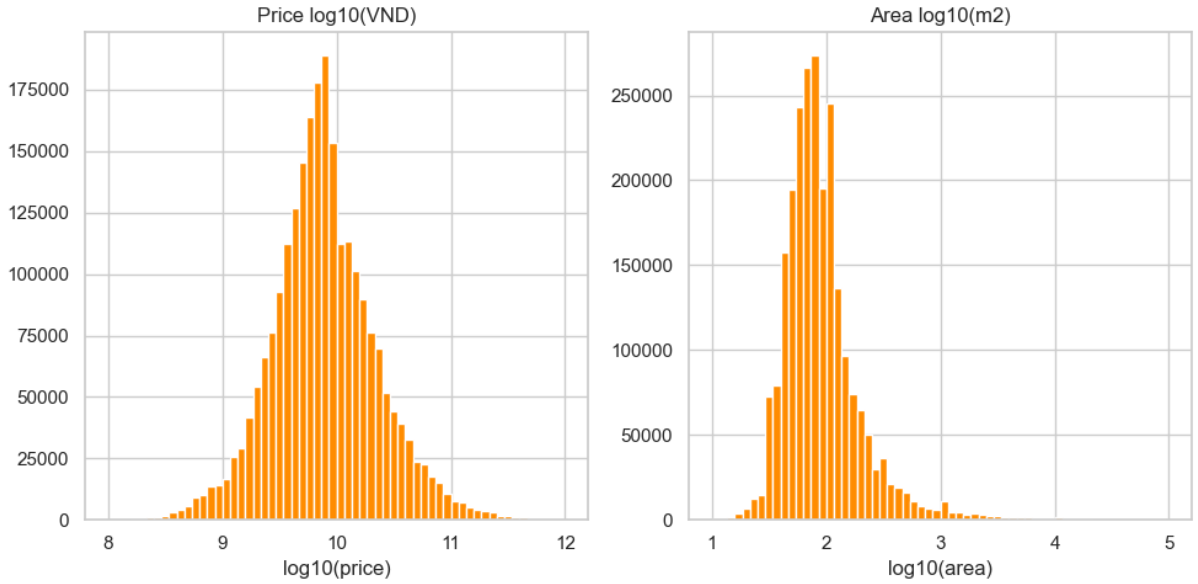


Figure 4: Histograms of $\log_{10}$(price) (left) and $\log_{10}$(area) (right).

Figure 4 shows that the distribution of price on the logarithmic scale is unimodal and approximately symmetric, with a mode at about 9.9 (roughly 8 billion VND) and values mainly between 8.5 and 11.5. The distribution of area is unimodal with a mode at about 1.9 to 2.0 (roughly 80 to 100 m^2) and a right tail that extends to about 5 (the upper limit of 100,000 m^2). Given the right skew shown in Table 5, this supports the use of a log transformation for price.

4.3.3 Group Coverage

The number of listings in each property type and province was counted. Groups with fewer than 100 listings are hard to use as features.

- Property types: all 5 types have more than 100 listings: houses (1,074,788), land (560,294), apartments (544,268), villas/townhouses (178,654), and shophouses (38,997).
- Provinces: 60 of the 63 provinces have at least 100 listings.
- Listings are concentrated in Hồ Chí Minh (928,687) and Hà Nội (776,760), which together account for about 71% of the analytical dataset.

4.3.4 Area–Price Relationship

The relationship between area and price was examined with a scatter plot on logarithmic axes, using a random sample of 20,000 listings (random seed 42) to limit overplotting.

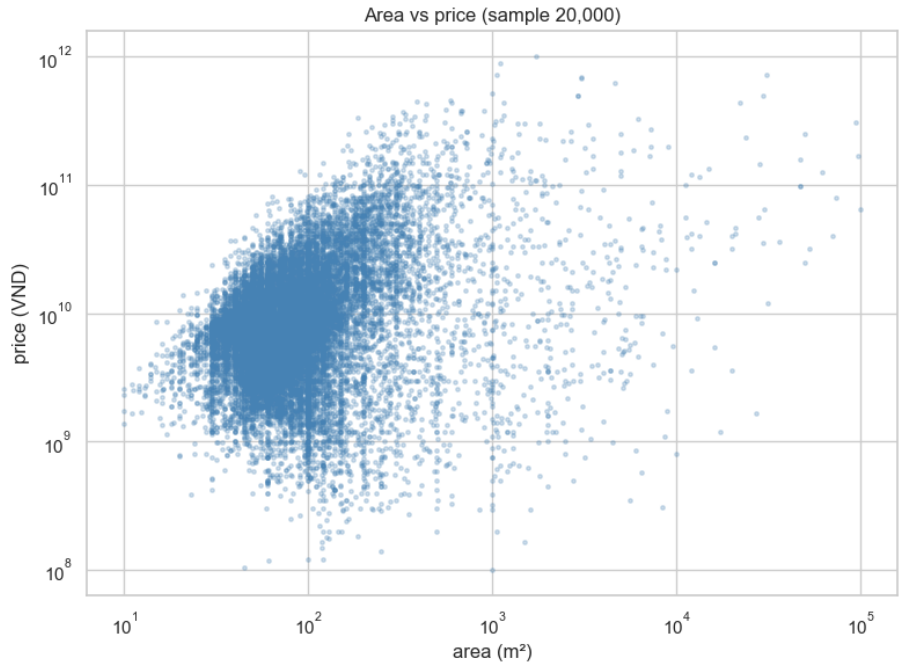


Figure 5: Area versus asking price on logarithmic axes (random sample of 20,000 listings).

Figure 5 shows that the area axis spans 10 to 10^5 m^2, the limits applied in the cleaning. Most points lie at areas of roughly 30 to 300 m^2 and prices of roughly 10^9 to 10^{10} VND. The cloud shows an upward tendency, so larger areas are generally associated with higher prices, but the dispersion at a given area is wide, and points become sparse above 10^3 m^2. This plot is a visual indication only; the strength of the association was not quantified.

4.3.5 Price per Square Metre by Property Type

The price per square metre (`price_per_m2`) is `price` divided by `area`. Its $\log_{10}$ distribution was plotted for each of the five property types, with the horizontal axis restricted to 6–10. A property type whose histogram has two peaks may mix different kinds of listing (for example, sale and rent).

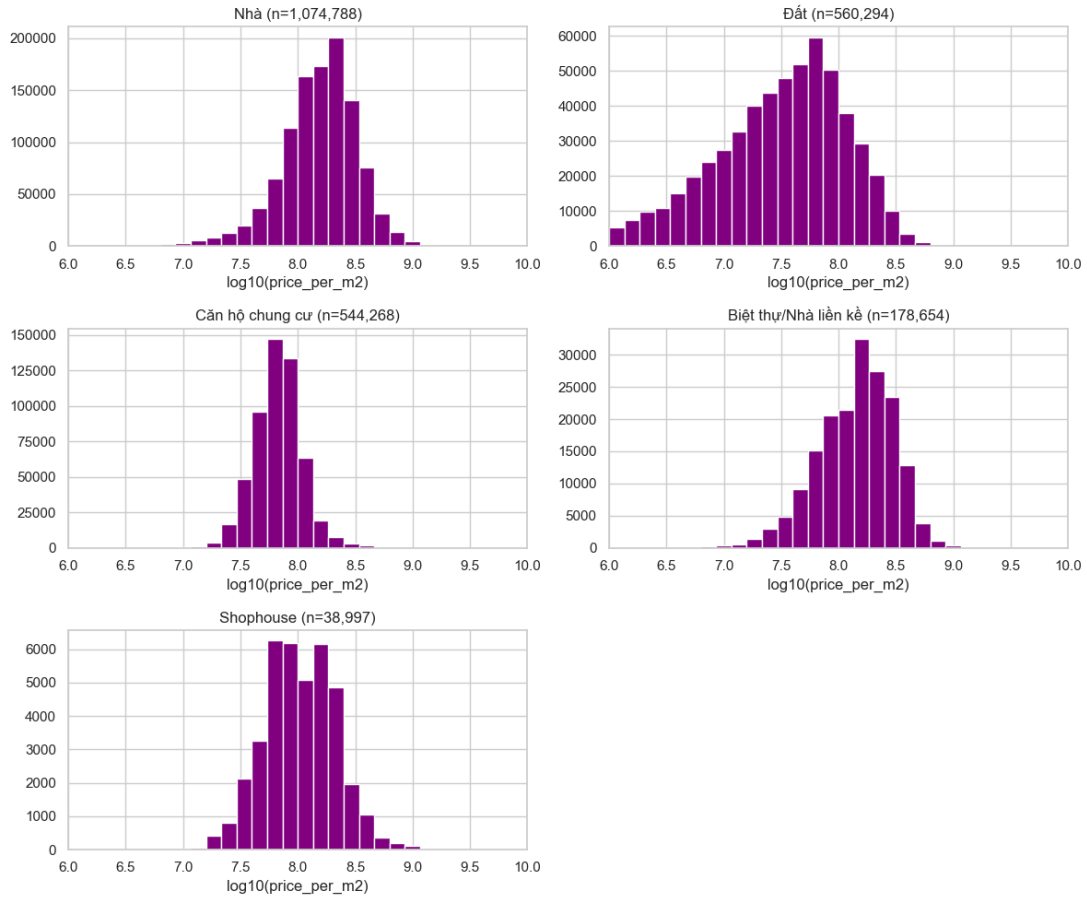


Figure 6: Distribution of $\log_{10}$(price per m^2) by property type (houses $n = 1{,}074{,}788$; land $n = 560{,}294$; apartments $n = 544{,}268$; villas/townhouses $n = 178{,}654$; shophouses $n = 38{,}997$).

Figure 6 shows that the distributions differ in location and spread across property types (the data are capped at 2 billion VND per m^2, i.e. $\log_{10} \approx 9.3$):

- Houses and villas/townhouses are centred at higher values (modes at about 8.3 and 8.2, roughly 200 and 160 million VND per m^2) than apartments and land (modes at about 7.8).
- Land has the widest distribution, skewed to the left, and apartments have the narrowest.
- Shophouses show two local maxima, at about 7.8–7.9 and 8.2. This may indicate a mix of different kinds of listing (for example, sale and rental), but it was not tested and remains a hypothesis.

4.4 Summary of Findings

- The raw data contain 3,500,744 listings. Missing values are frequent for structural attributes and depend on the property type.
- 24.72% of the raw listings were flagged as duplicates and 7.88% lack a valid price or area. After all filtering steps, 2,397,001 listings remain.
- Price and area are right-skewed, and the log transformation of price gives an approximately symmetric distribution. Extreme values were handled with fixed plausibility limits.
- All five property types and 60 of 63 provinces have sufficient listings, although listings are concentrated in two provinces. Area and price show a positive but widely dispersed relationship, and the price per square metre differs by property type.

References

[Tin26] TiniX AI. *Tinix Vietnam Real Estate Listings 2025–2026*. Hugging Face. 2026. URL: https://huggingface.co/datasets/tinixai/vietnam-real-estates (visited on 04/10/2026).